

Use Of Supercapacitors In Defence And Transport

Supercapacitors are energy storage devices that bridge the gap between electrolytic capacitors and rechargeable batteries. This article explains their importance and usage in defence and transportation sectors



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Electricity is a hugely versatile form of energy, but it suffers one major drawback; it is relatively difficult to store. Batteries can store large amount of power, but they take hours to charge up. Capacitors, on the other hand, charge almost instantly but store only tiny amounts of power.

In the next few decades, our fossil-fuelled cars and home heating appliances will need to be switched over to electric power, if we are to avert catastrophic climate change. Besides, in modern-day defence equipments, there is a need to supply high energy electric pulse. In our electric-powered future, where we will be required to store and release large amounts of electricity very quickly, it is quite likely we will turn to supercapacitors that combine the best of both the worlds.

What is a supercapacitor?

The structure of a capacitor is rather simple. It consists of two electric conductors called plates and an insulating layer between them called dielectric medium. If these two plates are connected to direct voltage, they become charged, and the magnitude of electric charge accumulated on the plates becomes proportional to the magnitude of voltage. The more charge it takes when voltage is applied, the higher the capacitance is. A supercapacitor is different from an ordinary capacitor mainly in two important ways—its plates effectively have a much bigger area and the distance between them is much smaller because the separator between them works in a different way than a conventional dielectric.

Like an ordinary capacitor, a supercapacitor also has two plates that are sepa-

rated. The plates are made of metal that is coated with a porous material like activated carbon, which provides the plates effectively with a bigger surface area, rendering it to store much more charge. For better understanding, let's assume electricity as water—where an ordinary capacitor is like a cloth that can mop up only a tiny little spillage of water, while a supercapacitor's porous plates can soak up many times more because they are more like a chunky sponge. Porous supercapacitors are big electricity sponges.

In an ordinary capacitor, the plates are separated by a relatively thick dielectric made of mica or plastic, or even simply air. When the capacitor is charged, positive charges form on one plate and negative charges on the other plate, creating an electric field between them. The field polarises the dielectric, so its molecules line up in the opposite direction to the field and reduce its strength. That means the plates can store more charge at a given voltage. This is shown in Fig. 1.

Classification of supercapacitors

Supercapacitors are mainly classified into three types: Double-layer capacitors; Pseudo-capacitors; Hybrid capacitors.

Double-layer capacitor (DLC). A capacitor consists of two electrodes, separator and electrolyte. As shown in Fig. 2, in a double-layer capacitor (DLC) there is no dielectric as such. Supercapacitors are often termed as DLC. The electrolyte in a DLC is the mixture of positive ions and negative ions dissolved in a solvent. Both the plates are soaked in an electrolyte and separated by a very thin insulator made of carbon, paper or plastic.

When the plates are charged up, an electrical double layer appears at the interface